Probabilistic Machine Learning

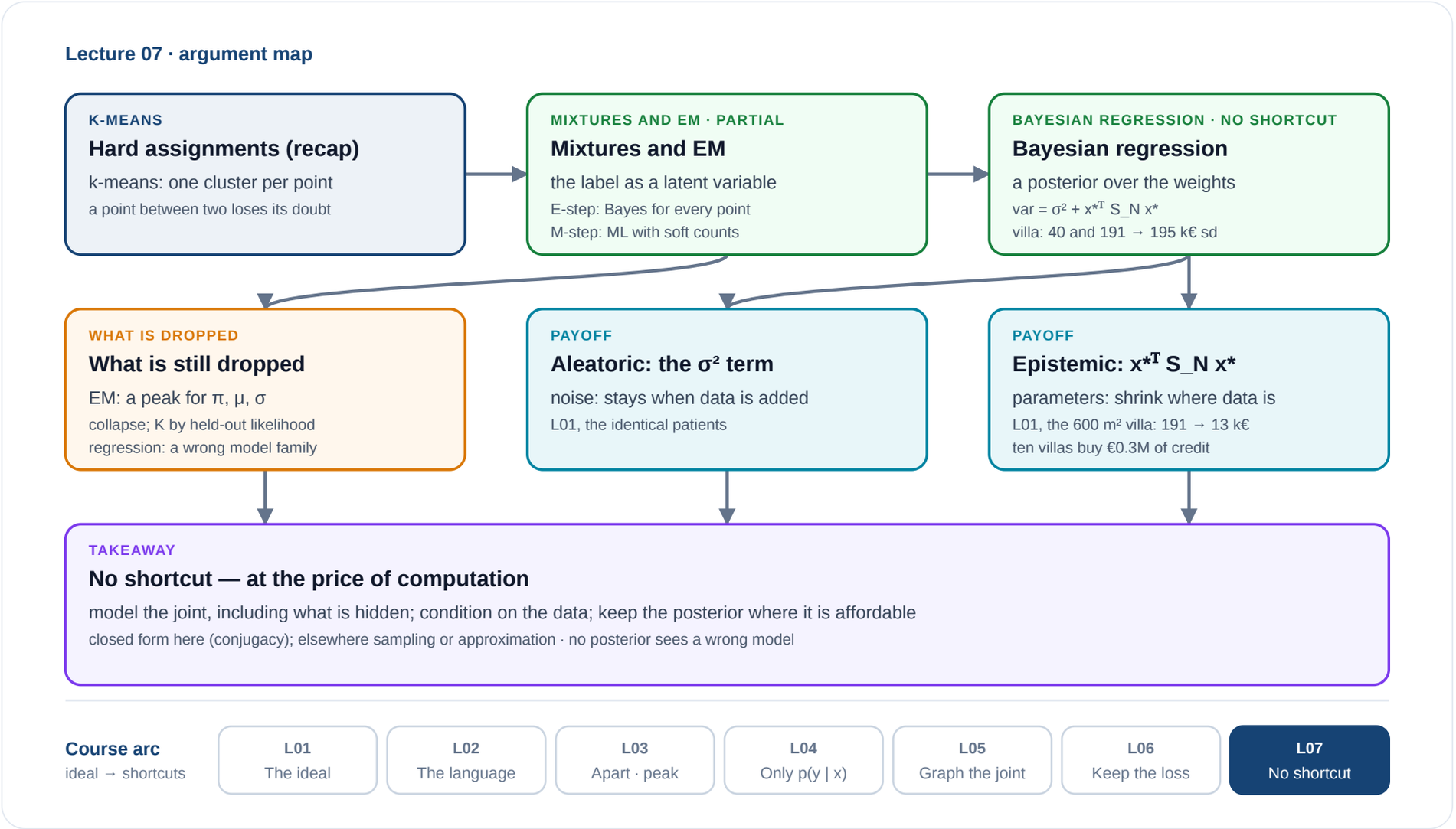
Lecture 07: Selected Probabilistic Algorithms

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Speaker notes

The lecture applies the procedure stated in Lecture 01 — model the joint distribution, condition on the observed data, read answers off the posterior — to two models. A Gaussian mixture treats the cluster label as a latent variable; EM computes the exact posterior of every label given the current parameters, but represents the parameters themselves by one maximum-likelihood point, so it is a partial shortcut. Bayesian linear regression with a Gaussian prior and known noise keeps the full Gaussian posterior over the weights; the variance of its predictive distribution is the sum of a noise term σ^2 and a parameter term $\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*$, evaluated on the house-price data of Lecture 01. k-means serves as the hard-assignment baseline: it is the limit of EM in which every posterior over a label collapses to 0 or 1.

Argument Map



Speaker notes

k-means assigns every point to exactly one cluster. A Gaussian mixture makes the assignment a latent variable z_i; EM conditions z_i on the data, but returns a single value of π, μ, σ, so it is exact only for the labels. Bayesian linear regression keeps the posterior over the weights $\mathbf{w}$, and its predictive variance is the sum of the noise term σ^2 (aleatoric: identical inputs, different outcomes) and the parameter term (epistemic: large far from the training data, such as the 600 m² villa of Lecture 01). Two limits remain even with the full posterior. Computing it is cheap only in conjugate cases; otherwise it needs sampling or approximation. And it quantifies uncertainty only within the assumed model family: a wrong family changes the predictive mean but leaves no trace in the predictive variance.

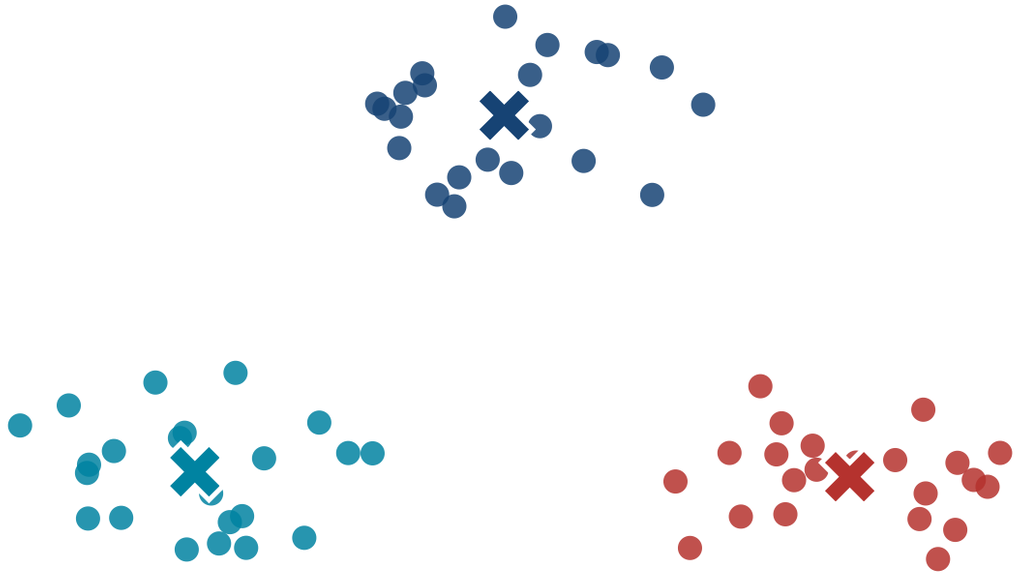
k-Means (Recap)

Guiding Question: k-means gives every point exactly one cluster. What is lost for a point that sits between two?

k-Means in One Slide

Minimise $\mathrm{SSE} = \sum_j \sum_{\mathbf{x} \in C_j} \|\mathbf{x} - \boldsymbol{\mu}_j\|^2$ by alternating: **assign** each point to the nearest centroid, **move** each centroid to the mean of its points. SSE never increases — it stops in a local optimum that depends on the start.

Move each centroid to the mean of its points (round 3)



Speaker notes

Each iteration has two half-steps. The assignment step gives each point to its nearest centroid, which for fixed centroids minimises SSE over the assignments. The update step moves each centroid to the mean of its points, which for fixed assignments minimises $\sum \|\mathbf{x} - \boldsymbol{\mu}_j\|^2$. Neither half-step can increase SSE and there are finitely many partitions, so the loop terminates — at a local optimum that depends on the initial centroids. Common remedies are several restarts and k-means++ seeding.

In the visualisation, the odd steps are assignment steps and the even steps update steps: three rounds on three well-separated blobs. The alternation "which cluster?" / "where is the cluster?" is the structure of EM's E-step and M-step in Lecture 07. The difference is that every k-means assignment is all-or-nothing, also for a point on the boundary between two clusters.

Where k-Means Fails

failure	what k-means assumes	fixed by a Gaussian mixture?
clusters of different spread	every cluster equally wide (distance only)	✓ each component has its own variance
elongated clusters	round clusters	✓ each component has its own covariance
clusters of different size	equal shares: the boundary lies halfway between centroids	✓ mixing weights π_k
a point between two clusters	one label per point	✓ a probability per cluster
rings, spirals	one blob per cluster	✗ still blobs — needs another representation
how many clusters	k is given	✗ still chosen — but by held-out likelihood

Speaker notes

The first four failures share one cause: k-means uses only the Euclidean distance to a centroid and gives each point one label. Its decision boundary between two clusters is therefore the perpendicular bisector of the two centroids. A Gaussian mixture with its own covariance Σ_k and weight π_k per component shifts the boundary towards the smaller or narrower component, and with unequal covariances makes it quadratic.

The last two rows are not fixed. Every component is still an ellipsoidal blob, so rings and spirals need a different representation (for example a similarity graph, as in spectral clustering) or many components per cluster. K remains a choice, but a mixture is a density, so K can be compared by held-out log-likelihood (or a penalised criterion such as BIC). For k-means the SSE, on training or held-out data, typically keeps decreasing as k grows, so it gives no comparable criterion.

A Point Between Two Clusters

A point sits exactly halfway between two centroids. k-means must answer "cluster 1" or "cluster 2". What should an algorithm say?

k-means minimises squared distance with **hard assignments**: fast, simple, and unable to say "I am not sure". The next module makes the label a latent variable — and k-means reappears as the limit where the doubt is removed.

Speaker notes

A point equidistant from two centroids contributes the same SSE under either label. k-means breaks the tie arbitrarily (in most implementations by index order) and reports the result with the same certainty as for a point lying on a centroid. Expressing "about half and half" requires a distribution over the label z, for example $P(z = 1 \mid \mathbf{x}) = P(z = 2 \mid \mathbf{x}) = 0.5$. In a mixture model this posterior is the responsibility; it equals 0.5 where the two weighted component densities are equal.

The lost information matters for decisions: an ambiguous case can be deferred — routed to a human, or held until more features are known — only if the ambiguity is represented. k-means is recovered from the mixture as the limit of equal, vanishing variances, in which every responsibility becomes 0 or 1.

A Point Between Two Clusters

A point sits exactly halfway between two centroids. k-means must answer "cluster 1" or "cluster 2". What should an algorithm say?

"About half and half" — which needs the cluster label to be a **random variable** with a distribution, not a value.

> k-means minimises squared distance with **hard assignments**: fast, simple, and unable to say "I am not sure". The next module makes the label a latent variable — and k-means reappears as the limit where the doubt is removed.

Mixture Models and EM

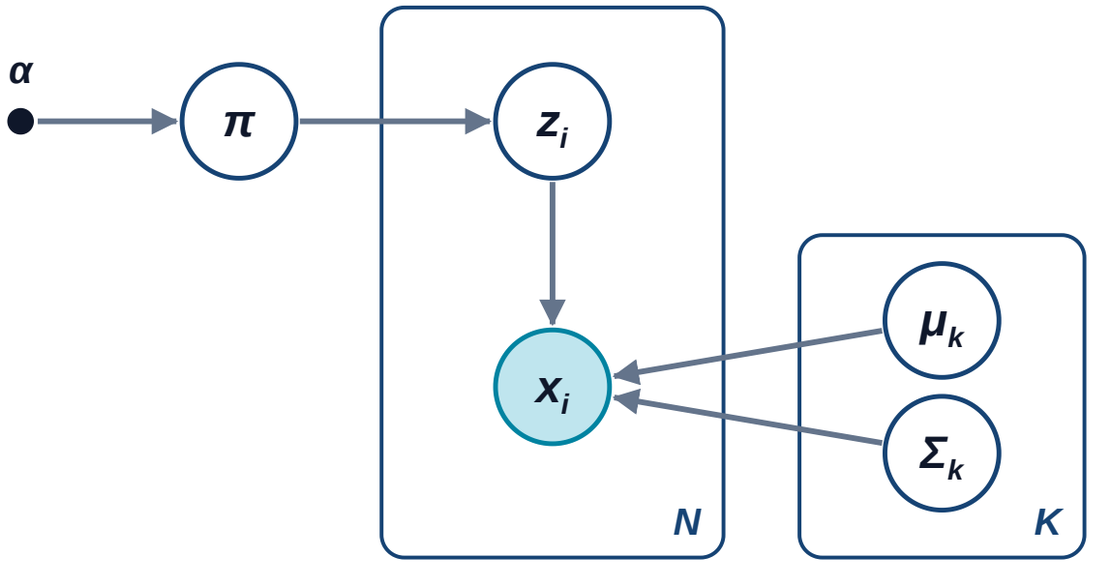
Guiding Question: If the cluster label were just another unobserved variable, could we infer it the way we infer anything else — and what would still be a guess?

The Label as a Hidden Variable

Thirty flats, described by their **price per m²**.
The listing does not say which district a flat is in — suburbs or old town.
A **Gaussian mixture** tells how such data arise:

1. pick a component $z \in \{1, \dots, K\}$ with probability π_z,
2. draw $x \sim \mathcal{N}(\mu_z, \sigma_z^2)$.

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2)$$



The plate of Lecture 05: z_i is **latent** — never observed.

Speaker notes

The generative story defines the joint $p(x, z) = \pi_z \mathcal{N}(x \mid \mu_z, \sigma_z^2)$; the density of x is its marginal, obtained by summing over z (Lecture 02). The weights satisfy $\pi_k \geq 0$ and $\sum_k \pi_k = 1$. In d dimensions σ_k^2 becomes a covariance matrix Σ_k, as drawn in the plate; one dimension keeps the formulas scalar.

The district is a genuine latent variable: every flat has one, the data set does not record it. The components are identifiable only up to relabelling — permuting the component indices leaves $p(x)$ unchanged — so the likelihood has $K!$ equivalent maxima, and "component 1" has no meaning beyond a given fit.

Why There Is No Closed Form

For one Gaussian, maximum likelihood — the recipe of Lecture 03 — gives the sample mean and variance. For a mixture, the log-likelihood is a **log of a sum**:

$$\ell(\theta) = \sum_{i=1}^{N} \log \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$$

The sum inside the log ties all components together; setting the gradient to zero gives no formula.

If you knew each flat's district, how would you fit the two Gaussians? If you knew the two Gaussians, how would you find each flat's district?

Speaker notes

For a single Gaussian, $\log \mathcal{N}$ is quadratic in μ, and setting the derivatives to zero gives the sample mean and variance. For the mixture, the derivative with respect to μ_k is $\sum_i \gamma_{ik}(x_i - \mu_k)/\sigma_k^2$, where γ_{ik} (next slide) itself depends on all parameters. The stationarity condition $\mu_k = \sum_i \gamma_{ik} x_i / \sum_i \gamma_{ik}$ is an implicit equation, not a formula.

With known labels, the complete-data log-likelihood $\sum_i \log\left(\pi_{z_i} \mathcal{N}(x_i \mid \mu_{z_i}, \sigma_{z_i}^2)\right)$ has the log directly on each term and splits into one problem per group: π_k is the share of points in group k, μ_k and σ_k^2 are the mean and variance of that group. With known parameters, each label's posterior follows from Bayes' theorem. EM alternates the two; its M-step formulas are exactly the stationarity conditions above with γ held fixed.

Why There Is No Closed Form

For one Gaussian, maximum likelihood — the recipe of Lecture 03 — gives the sample mean and variance. For a mixture, the log-likelihood is a **log of a sum**:

$$\ell(\theta) = \sum_{i=1}^{N} \log \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$$

The sum inside the log ties all components together; setting the gradient to zero gives no formula.

If you knew each flat's district, how would you fit the two Gaussians? If you knew the two Gaussians, how would you find each flat's district?

Known labels: count, average and take the spread per district — maximum likelihood per group. Known Gaussians: Bayes' theorem per flat. **EM alternates the two.**

The E-Step: Bayes for Every Point

Given the current parameters, the **responsibility** of component k for point i is the posterior of the hidden label:

$$\gamma_{ik} = P(z_i = k \mid x_i) = \frac{\pi_k \cdot \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}$$

Prior (the share of component k) times likelihood (how well k explains x_i), normalised over the components.

> The same computation as Lecture 02's disease test — once per data point. A flat halfway between the two districts gets about $0.5/0.5$.

Speaker notes

γ_{ik} is the posterior $P(z_i = k \mid x_i, \theta)$ under the current parameters θ; the denominator is the mixture density $p(x_i)$. For each point the responsibilities are non-negative and sum to 1 over k. The E-step keeps the whole distribution over each label instead of its most probable value, and given θ it is exact — this is the part of EM that conditions on the data as the ideal prescribes.

The structure is that of the disease test in Lecture 02: π_k is the base rate, the component density the likelihood. A point gets $0.5/0.5$ where the two weighted densities are equal; with unequal weights or variances this point is in general not the midpoint between the means. Implementations compute γ_{ik} in log space (log-sum-exp), because the densities underflow for points far from a component.

The M-Step: Maximum Likelihood With Soft Counts

$$N_k = \sum_i \gamma_{ik}, \qquad \pi_k = \frac{N_k}{N}$$

$$\mu_k = \frac{1}{N_k} \sum_i \gamma_{ik} x_i, \qquad \sigma_k^2 = \frac{1}{N_k} \sum_i \gamma_{ik} (x_i - \mu_k)^2$$

Expectation–Maximisation

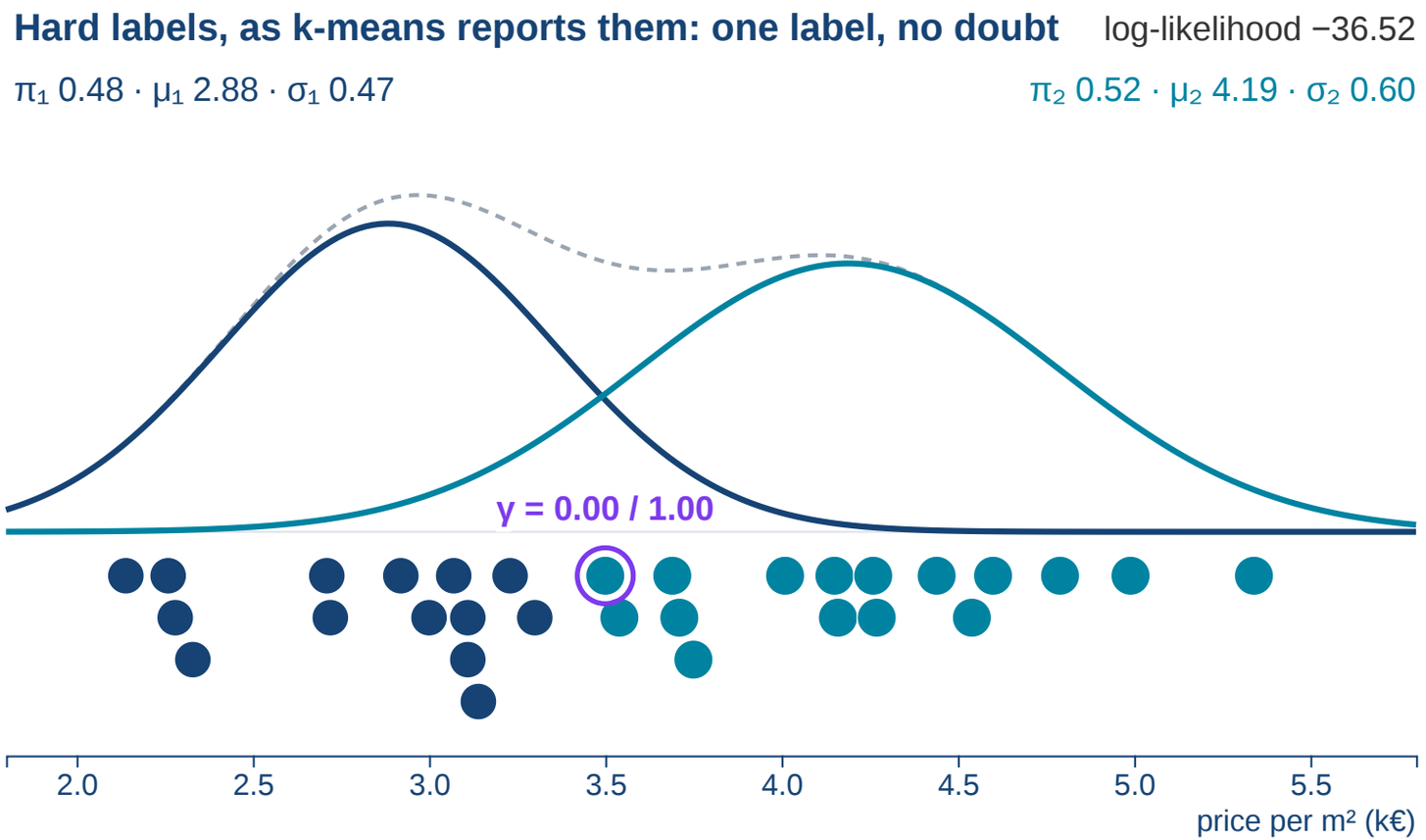
1. Start with a guess for π_k, μ_k, σ_k.
2. **E-step:** compute every γ_{ik} (Bayes).
3. **M-step:** refit π_k, μ_k, σ_k — each point counts γ_{ik} times.
4. Repeat until $\ell(\theta)$ stops rising — it never decreases.

Speaker notes

With γ fixed, the M-step maximises the expected complete-data log-likelihood $\sum_i \sum_k \gamma_{ik} \left[\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \sigma_k^2) \right]$ — maximum likelihood (Lecture 03) with each point counted fractionally in every component. N_k is the effective number of points in component k. The update for π_k follows with a Lagrange multiplier for $\sum_k \pi_k = 1$; the variance is divided by N_k, so it is the (biased) maximum-likelihood estimate. With $\gamma_{ik} \in \{0, 1\}$ the formulas reduce to fitting each group separately.

Monotonicity: the E-step makes a lower bound on $\ell(\theta)$ tight at the current θ, and the M-step raises that bound, so ℓ cannot decrease (Dempster, Laird and Rubin, 1977). EM converges to a stationary point, usually a local maximum that depends on the start; several restarts or an initialisation from k-means are common practice.

Watching EM



Predict before the first click: which flats will come out half-coloured — and does the flat at 3.5 k€/m² end up decided?

Speaker notes

Data: 30 flats with prices between 2.14 and 5.34 k€/m². Start: two components placed poorly, $\mu = 2.2$ and 3.4, $\sigma = 0.6$, $\pi = 0.5/0.5$; log-likelihood −51.6. Step 1 (E): every flat is coloured by its responsibilities (navy = component 1, teal = component 2). Step 2 (M): the refit gives $\pi \approx 0.24/0.76$ and $\mu \approx 2.65/3.85$; the log-likelihood rises to −36.9. Steps 3–4: the second round changes little (−36.8). Step 5: after 60 iterations $\pi \approx 0.48/0.52$, $\mu \approx 2.89/4.19$, $\sigma \approx 0.47/0.60$, log-likelihood −36.52. Step 6: the same fit with every responsibility rounded to 0 or 1, as k-means would report it.

At convergence the flats between about 3.2 and 3.8 k€/m² have responsibilities between 0.2 and 0.8. The ringed flat at 3.5 has $\gamma \approx 0.49/0.51$ — it is not decided; in step 6 it carries a single label with no indication that it is a near tie.

EM and k-Means

	k-means	EM on a Gaussian mixture
assignment	hard: one cluster per point	soft: responsibilities γ_{ik}
cluster shape	equal and round (distance only)	own variance (covariance) and weight
objective	SSE	log-likelihood $\ell(\theta)$
output	labels and centroids	a density $p(x)$ and a posterior per point

k-means is the limit of EM with equal, spherical components whose variance shrinks to zero: the responsibilities collapse to 0 or 1, and the M-step becomes "move to the mean".

Speaker notes

With equal weights and spherical covariances $\sigma^2 I$, $\gamma_{ik} \propto \exp\left(-\|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 / 2\sigma^2\right)$. As $\sigma \to 0$ the term with the smallest distance dominates every other one exponentially, so $\gamma_{ik} \to 1$ for the nearest centroid and 0 otherwise; the M-step mean becomes the mean of the assigned points. For this reason k-means is also called hard EM. It shares EM's dependence on the start, but optimises SSE rather than $\ell(\theta)$, and has no variances or weights that could shift the boundary between clusters.

What EM Still Drops

✓ **The hidden labels:** a full posterior per point — the ideal, carried out.

✗ **The parameters:** π_k, μ_k, σ_k are **one point estimate** — Lecture 03's peak. The plate draws them as unknowns; EM gives them no error bar.

✗ **A peak that may not exist:** a component that shrinks onto one point has $\sigma_k \to 0$ and infinite likelihood. In practice: a floor on σ_k, or a prior (MAP).

✗ **The number of components:** the training likelihood always rises with K. Choose K by the **held-out** log-likelihood (Lecture 06), not by an elbow.

> EM is a **partial shortcut**: Bayes for what is hidden, a peak for the parameters. Keeping a posterior over π, μ, σ as well needs sampling or variational inference.

Speaker notes

EM conditions the labels on the data but returns one value of π, μ, σ; parameter uncertainty is not represented, although the plate treats the parameters as unknown. The course roadmap's "no shortcut" row for this lecture carries exactly this qualification: EM is the ideal only for the hidden labels.

Degeneracy: with $\mu_k = x_i$ and $\sigma_k \to 0$, the term $\pi_k \mathcal{N}(x_i \mid x_i, \sigma_k^2) = \pi_k / (\sqrt{2\pi}\,\sigma_k)$ diverges while the other points keep a finite density from the other components, so the likelihood is unbounded and its "maximum" is a spike on one point. Libraries therefore add a small constant to the covariance (in scikit-learn's `GaussianMixture`, `reg_covar`) or place a prior on it (MAP-EM).

Number of components: a mixture with $K+1$ components contains the one with K (set $\pi_{K+1} = 0$), so the maximised training likelihood cannot fall as K grows. The held-out log-likelihood is the log-loss of Lecture 06 applied to a density; BIC is a common penalised alternative.

Mini-Exercise

Component 1 has weight $\pi_1 = 0.75$, component 2 has $\pi_2 = 0.25$. For one flat, $\mathcal{N}(x \mid \mu_1, \sigma_1^2) = 0.2$ and $\mathcal{N}(x \mid \mu_2, \sigma_2^2) = 0.6$.

What are its responsibilities?

Speaker notes

Weighted likelihoods: $0.75 \cdot 0.2 = 0.15$ and $0.25 \cdot 0.6 = 0.15$; their sum $p(x) = 0.30$ is the normaliser, so $\gamma = 0.15/0.30 = 0.5$ for each component. In odds form: prior odds 3 : 1 for component 1 times likelihood ratio 1 : 3 gives posterior odds 1 : 1 — the base-rate effect of Lecture 02, for a cluster label. If both components have the same σ, the larger density means that x is closer to μ_2, so nearest-centroid assignment would give the flat to component 2 outright. The values 0.2 and 0.6 are densities, not probabilities (a narrow component can exceed 1); only their ratio enters γ.

Mini-Exercise

Component 1 has weight $\pi_1 = 0.75$, component 2 has $\pi_2 = 0.25$. For one flat, $\mathcal{N}(x \mid \mu_1, \sigma_1^2) = 0.2$ and $\mathcal{N}(x \mid \mu_2, \sigma_2^2) = 0.6$.

What are its responsibilities?

$0.75 \cdot 0.2 = 0.15$ and $0.25 \cdot 0.6 = 0.15$, so $\gamma = 0.5/0.5$. The prior favours component 1 three to one, the likelihood favours component 2 three to one — they cancel. Nearest-centroid assignment would ignore the weights.

Takeaway

A mixture model makes the cluster label a **latent variable with a posterior**. EM alternates **Bayes for the labels** (E) and **weighted maximum likelihood for the parameters** (M): the posterior over what is hidden, a peak for the rest. k-means is what remains when the doubt is squeezed out.

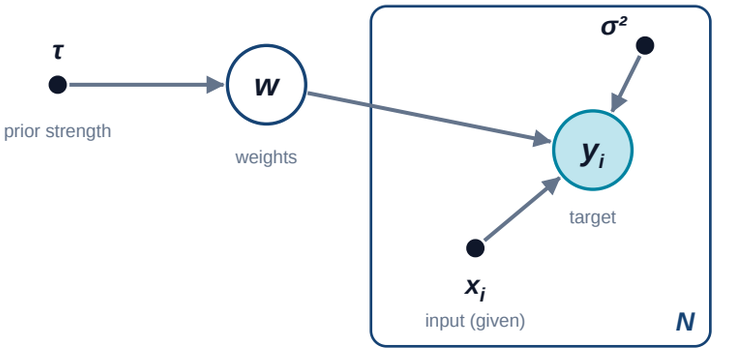
Bayesian Linear Regression

Guiding Question: Your model prices a 600 m² villa at €2.1M — five times larger than any flat it was trained on. Should it say how sure it is, and how much of its doubt could data remove?

Keeping the Distribution

Lecture 01's house-price model: **20 flats of 40–120 m²**, and a question about a **600 m² villa**. The model and the plate of Lecture 05:

$$y_i = w_0 + w_1 x_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$



- → **Likelihood:** the Gaussian noise model of Lecture 04, $\sigma = 40$ k€.
- → **Prior:** $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_0)$ — intercept within about ±1,000 k€, slope within about ±20 k€/m². Weak.
- → **Posterior:** Gaussian again — the prior is **conjugate**.

Speaker notes

Maximum likelihood (Lecture 03, Lecture 04) returns the $\mathbf{w}$ that maximises $p(D \mid \mathbf{w})$; MAP returns the maximiser of the posterior $p(\mathbf{w} \mid D)$. Bayesian regression keeps the whole posterior — the two-parameter analogue of the Beta posterior in Lecture 03.

Prior: $\mathbf{S}_0 = \mathrm{diag}(500^2, 10^2)$, intercept in k€, slope in k€/m². Two prior standard deviations are ±1,000 k€ and ±20 k€/m² — wide compared with the data-supported slope of about 3.5 k€/m², so the data dominate. Conjugacy: with a linear model, Gaussian noise and known σ, the log-likelihood is quadratic in $\mathbf{w}$, as is the log of a Gaussian prior; their sum is quadratic again, so the posterior is Gaussian. With σ unknown and a normal–inverse-gamma prior the posterior is still in closed form, and the predictive distribution is a Student t — heavier-tailed, and with 20 observations only slightly wider.

Precision Adds Up

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}, \qquad \mathbf{m}_N = \frac{1}{\sigma^2}\mathbf{S}_N\mathbf{X}^\top\mathbf{y}$$

The update is adding

Posterior precision = prior precision + data precision — Lecture 03's **adding counts**, for Gaussians. Every flat adds $\frac{1}{\sigma^2}\mathbf{x}_i\mathbf{x}_i^\top$.

The centre is the ridge estimate

$\mathbf{m}_N$ is the MAP estimate — ridge regression. With this weak prior it is almost least squares: slope **3.496** vs **3.501** k€/m².

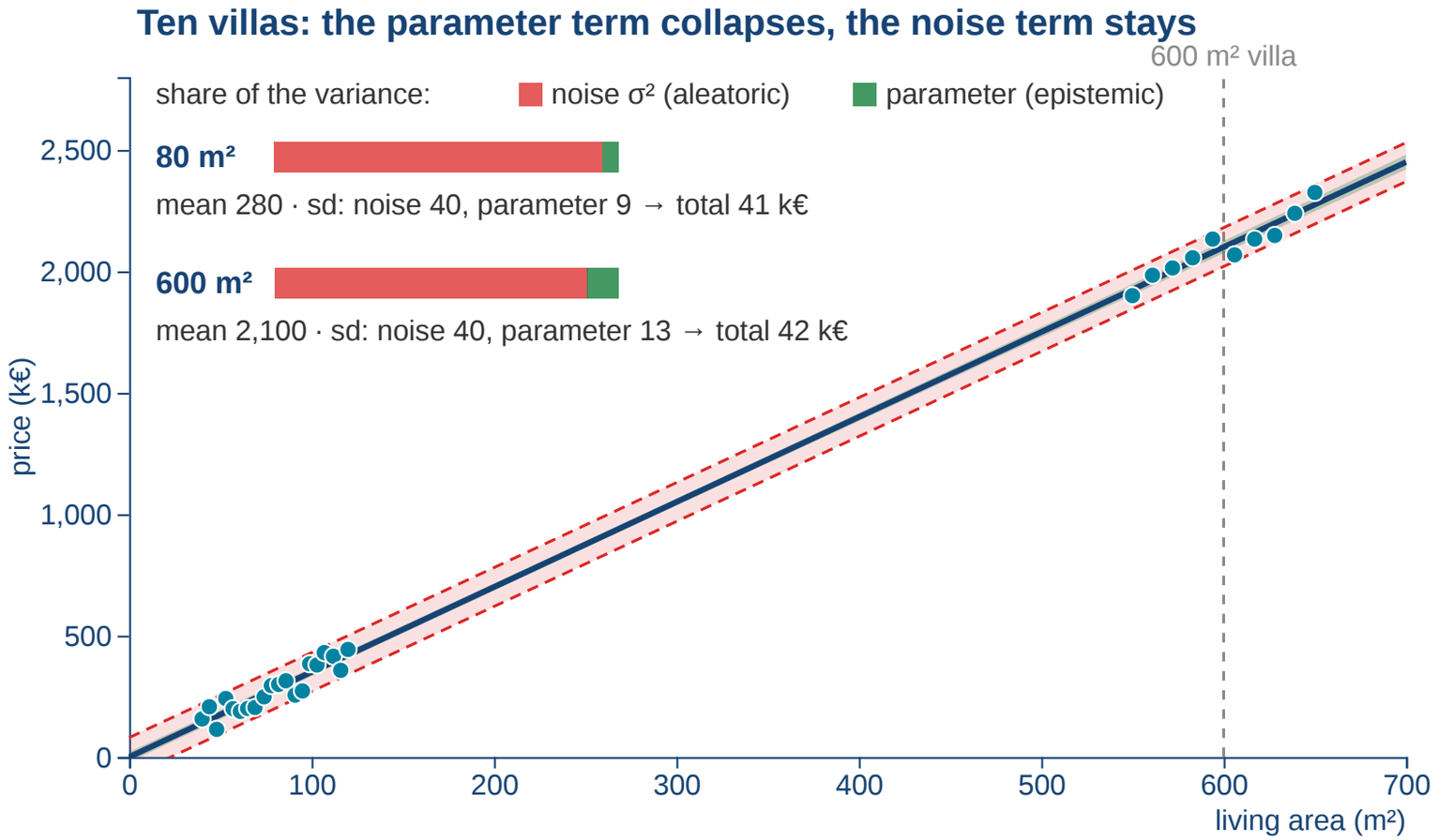
What is new is not $\mathbf{m}_N$ but $\mathbf{S}_N$: the slope is 3.50 ± 0.37 k€/m² (one sd) — and that uncertainty is multiplied by 600 m² at the villa.

Speaker notes

Derivation: the log-posterior is $-\frac{1}{2\sigma^2}\|\mathbf{y}-\mathbf{X}\mathbf{w}\|^2 - \frac{1}{2}\mathbf{w}^\top\mathbf{S}_0^{-1}\mathbf{w}$ plus a constant. The quadratic term in $\mathbf{w}$ gives the precision $\mathbf{S}_N^{-1}$, the linear term gives $\mathbf{S}_N^{-1}\mathbf{m}_N = \mathbf{X}^\top\mathbf{y}/\sigma^2$ (zero prior mean). $\mathbf{X}$ has rows $(1, x_i)$, so every flat adds $\mathbf{x}_i\mathbf{x}_i^\top/\sigma^2$ to the precision — the Gaussian analogue of Beta$(a+h, b+n-h)$.

$\mathbf{m}_N$ is the ridge solution with penalty $\sigma^2\mathbf{S}_0^{-1}$, here a different penalty per coefficient ($40^2/500^2 = 0.0064$ on the intercept, $40^2/10^2 = 16$ on the slope). Posterior mean: intercept 0.3 k€, slope 3.496 k€/m²; least squares: −0.07 k€ and 3.501 k€/m². A weak prior barely moves the line; what is new is $\mathbf{S}_N$. Posterior sds: 30.7 k€ for the intercept, 0.37 k€/m² for the slope, correlation −0.96. All flats lie far from $x = 0$ (40–120 m², mean 80 m²), so a steeper line fits only with a lower intercept: plausible lines pivot about the centre of the data, and the parameter sd $\sqrt{\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*}$ grows with the distance from it — 191 k€ at 600 m².

Watching the Band Form



Predict before the last click: ten villas of 550–650 m² join the data. Which part of the band shrinks — and which part cannot?

Speaker notes

Step 0: the 20 flats (40–120 m²) and the query at 600 m². Step 1: keep the peak (Lecture 04) — the least-squares line with the plug-in band ±2σ = ±80 k€, equally wide everywhere, because uncertainty about $\mathbf{w}$ is ignored. Step 2: fifteen lines drawn from $\mathcal{N}(\mathbf{m}_N, \mathbf{S}_N)$; they agree on the flats and fan out towards 600 m², since the slope uncertainty (sd 0.37 k€/m²) is multiplied by the distance from the data. Step 3: the predictive band (±2 sd) split into its two parts; the bars show variance shares — at 80 m² the noise is 95% (1,600 of 1,680 k€²), at 600 m² the parameter term is 96% (36,419 of 38,019 k€²). Step 4: ten villas of 550–650 m² join the data; the parameter sd at 600 m² falls from 191 to 13 k€, while the noise part — the dashed red lines at ±2σ around the mean — does not move.

The parameter part shrinks with data near the query; the noise part cannot, because σ describes the spread of prices at a given size, not a lack of data.

Two Terms of One Prediction

$$p(y^* \mid x^*, D) = \mathcal{N}\left(\mathbf{m}_N^\top \mathbf{x}^*,\ \ \sigma^2 + \mathbf{x}^{*\top} \mathbf{S}_N \mathbf{x}^*\right)$$

■ noise term — aleatoric, fixed · ■ parameter term — epistemic, shrinks with data

query	mean	noise sd	parameter sd	total sd
80 m² (inside the data)	280 k€	40	9	**41**
300 m²	1,049 k€	40	81	**90**
600 m² — the villa	2,098 k€	40	191	**195**
600 m², after ten villas	2,100 k€	40	13	**42**

> Near the data the **noise** dominates, at the villa the **ignorance**. Ten villas remove the ignorance; the noise stays — Lecture 01's two examples.

Speaker notes

Derivation: $y^* = \mathbf{w}^\top \mathbf{x}^* + \varepsilon^*$ with $\mathbf{w} \sim \mathcal{N}(\mathbf{m}_N, \mathbf{S}_N)$ independent of $\varepsilon^* \sim \mathcal{N}(0, \sigma^2)$. A linear function of a Gaussian plus independent Gaussian noise is Gaussian, with mean $\mathbf{m}_N^\top \mathbf{x}^*$ and variance $\mathbf{x}^{*\top} \mathbf{S}_N \mathbf{x}^* + \sigma^2$.

Variances add, standard deviations do not: at 600 m², $40^2 + 191^2 = 1{,}600 + 36{,}419 = 38{,}019 \approx 195^2$. A small term therefore vanishes next to a large one: 9 k€ of parameter sd next to 40 k€ of noise adds 1 k€ to the total. At 120 m², the edge of the data: parameter sd 17, total 44 k€. All numbers use $\sigma = 40$ k€ known and $\mathbf{S}_0 = \mathrm{diag}(500^2, 10^2)$; the flat-prior limit gives the same values to the precision shown.

The parameter variance falls like $1/n$ when the design is repeated, and grows with the squared distance from the data. A thousand more flats spread over 40–120 m² would cut the parameter sd at 600 m² from 191 to about 28 k€ — still more than the 13 k€ that ten villas achieve. The noise term stays at 40 k€ in both cases.

What the Posterior Cannot Know

The band is uncertainty **within the linear model**. Suppose villas actually sell at 5,000 €/m²: the villa is worth **€3.0M** — far outside 2,098 ± 2 · 195 k€, i.e. €1.71M to €2.49M.

The world is noisy

shows up in σ^2, the noise term — accept it, or measure new attributes

Too few data here

shows up in $\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*$, the parameter term — collect data there

The family is wrong

shows up **nowhere** in the formula — check on held-out data where you predict; compare model families

A wide band is an honest "I don't know". A narrow band is not a proof — it can be confidently wrong if the straight line itself is wrong.

Speaker notes

Under the linear model the 95% predictive interval at 600 m² is $2{,}098 \pm 390$ k€, i.e. 1,708 to 2,488 k€. A true price of 5,000 €/m² · 600 m² = 3,000 k€ lies 902 k€ above the mean, about 4.6 predictive standard deviations. Both terms of the predictive variance are computed conditional on the model family (linear mean, Gaussian noise of constant variance). Misspecification shifts the mean, but nothing in the variance reflects it.

It can be detected only from outside the posterior: by held-out data near the query (residuals far outside the band), or by comparing model families — for example a quadratic term or a log-price model — on held-out log-likelihood or marginal likelihood. The three cards are the three answers of the reasoning cycle in Lecture 01 ("noise, too little data, or the wrong model?"): noise appears in σ^2, too little data in $\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*$, a wrong model in no term.

Deciding With the Predictive Distribution

A lender lends up to the value it is about 98% sure the property exceeds: the mean minus two standard deviations.

€198k

an 80 m² flat: $280 - 2 \cdot 41$

€1.71M

the villa today: $2{,}098 - 2 \cdot 195$

€2.02M

after ten villa sales: $2{,}100 - 2 \cdot 42$

Ten comparable sales are worth about **€0.3M of credit**: buying data pays where the **parameter** term dominates. Near the data it buys nothing — the €40k noise stays.

Speaker notes

The lender's rule reads a quantile of the predictive distribution, so both variance terms enter the decision. For a Gaussian, mean − 2 sd is exceeded with probability 0.977 ("about 98%" on the slide); the exact 97.5% quantile would be mean − 1.96 sd. Amounts: $280 - 2 \cdot 41 = 198$ k€; $2{,}098 - 2 \cdot 195 = 1{,}708$ k€; $2{,}100 - 2 \cdot 42 = 2{,}016$ k€. Ten villa sales raise the credit by 308 k€.

The same calculation values information. Data reduce only the parameter term, so they pay where that term dominates. At 80 m² a thousand more flats would cut the total sd from 41.0 to about 40.0 k€ and raise the loan by about 2 k€. A lender using the plug-in band of maximum likelihood (±2σ = ±80 k€) would lend about 2,018 k€ on the villa today — an amount the Bayesian model supports only after the ten villa sales; the unrepresented parameter sd of 191 k€ is the exposure. This is the simplest case of keeping uncertainty up to the decision, as set out in Lecture 01.

Mini-Exercise

Model A (maximum likelihood with the plug-in noise of Lecture 04) and model B (Bayesian) both predict about **€2.1M** for the 600 m² villa. A reports **±80 k€**.

What does B report — and which one do you show the lender?

Speaker notes

B's total sd at 600 m² is 195 k€, so ±2 sd is ±390 k€. A's ±80 k€ is the plug-in interval $\mathcal{N}(\hat{\mathbf{w}}^\top \mathbf{x}^*, \sigma^2)$ of Lecture 04, which treats $\hat{\mathbf{w}}$ as exact; at the villa it is as narrow as B's interval for an 80 m² flat (±82 k€).

The classical frequentist prediction interval, $\hat{y} \pm t_{0.975,\,n-2}\, s\sqrt{1 + \mathbf{x}^{*\top}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}^*}$, also contains a parameter term; here $s = 40.0$ k€, $t_{0.975,18} = 2.10$, and the half-width is about 410 k€. The widening is therefore not specific to Bayesian methods; what is Bayesian is reading the interval as a probability statement about this villa's price given the data. The centres agree: B's mean is the ridge estimate, which with this weak prior differs from least squares by about 2.5 k€ at 600 m² (2,098 vs. 2,100.5 k€).

Mini-Exercise

Model A (maximum likelihood with the plug-in noise of Lecture 04) and model B (Bayesian) both predict about **€2.1M** for the 600 m² villa. A reports **±80 k€**.

What does B report — and which one do you show the lender?

B reports about **±390 k€** ($2 \cdot 195$): its parameter term is large this far from the data. Show **B** — not because its centre is better (both are ≈ €2.1M) but because A claims the precision of an 80 m² flat for a house it has never seen anything like.

Takeaway

Bayesian regression keeps a **posterior over the weights** and predicts with a **predictive distribution** whose variance has two terms: the **noise** term (aleatoric) that no data removes, and the **parameter** term (epistemic) that grows off the data and shrinks where data is collected. What neither term covers is a wrong model family.

Course Synthesis

Guiding Question: Seven lectures, one ideal, several shortcuts. Which questions can each method you know still answer — and what would it cost to answer the rest?

The Shortcuts, Revisited

Lecture 01's table, with what the course has learned about each row:

shortcut	what it dropped	bought back by	at the cost of
Keep the peak · 03–04	how sure we are about the parameters	the posterior (Bayesian Linear Regression)	a prior, an integral
Assume it apart · 03	the dependences assumed away	a graph that draws them	more parameters
Only $p(y \mid x)$ · 04	every question about x	a generative model	modelling x
Graph the joint · 05	the edges not drawn	more edges	costlier inference
Keep the loss · 06	every probability	calibration, log-loss, error bars	held-out data
EM, partial · 07	how sure we are about π, μ, σ	priors, sampling or VI	computation
No shortcut · 07	nothing — within the model family	—	a prior; the family itself

> Every shortcut is a trade of an answer for computation or data. Knowing which one a method takes tells you which question it can no longer answer.

Speaker notes

The table extends the shortcut table of Lecture 01 by what each shortcut discards and what restores it. Keeping the peak (maximum likelihood, MAP) discards the spread of the parameter posterior; the full posterior restores it at the cost of a prior and an integral over the parameters. Assuming independence discards dependences; a graph can encode them, at the cost of more parameters. Modelling only $p(y \mid x)$ makes every question about x (missing features, outliers, generating data) unanswerable without a model of x. A graph of the joint is only as good as its missing edges; adding edges makes inference costlier. A pure loss discards probabilities; calibration, log-loss and error bars recover them, using held-out data.

EM is listed separately because it is exact for the hidden labels (the E-step is Bayes' theorem) but a point estimate for π, μ, σ; a posterior over those needs MCMC or variational inference. The "no shortcut" row is exact only conditional on the linear-Gaussian model family and the chosen prior. The common reason for every shortcut is cost: integrals over parameters or latent variables are replaced by optimisation, or by more data.

What Comes After

- → **Gaussian processes** — Bayesian regression with infinitely many basis functions: the two-term band of Bayesian linear regression, without the straight line.
- → **Bayesian neural networks and deep ensembles** — the parameter term for models with millions of weights, approximately.
- → **Sampling (MCMC) and variational inference** — conditioning on the data when there is no closed form; EM is the special case that keeps a posterior only for the hidden labels.

> Model the joint, condition on the data, read off answers — and when you take a shortcut, **name it and know what it drops**. A prediction worth acting on says how sure it is, and why: noise, ignorance, or a model that may be wrong.

Speaker notes

Gaussian processes: Bayesian regression on basis functions $\phi(x)$ with prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_0)$ implies the prior covariance $\phi(x)^\top \mathbf{S}_0 \phi(x')$ between function values. Replacing it by a kernel $k(x, x')$ corresponds, for kernels such as the squared exponential, to infinitely many basis functions. The predictive variance is again noise plus a term that grows away from the data.

Bayesian neural networks put a prior on the weights; the posterior has no closed form and is approximated (Laplace approximation, variational inference, MCMC). Deep ensembles — several networks trained from different random initialisations — are a common practical approximation of the parameter term.

MCMC draws samples whose distribution converges to the posterior. Variational inference chooses the member of a tractable family that maximises the evidence lower bound (ELBO). EM maximises the same bound, with the latent-variable posterior computed exactly in the E-step and the parameters optimised to a point in the M-step.